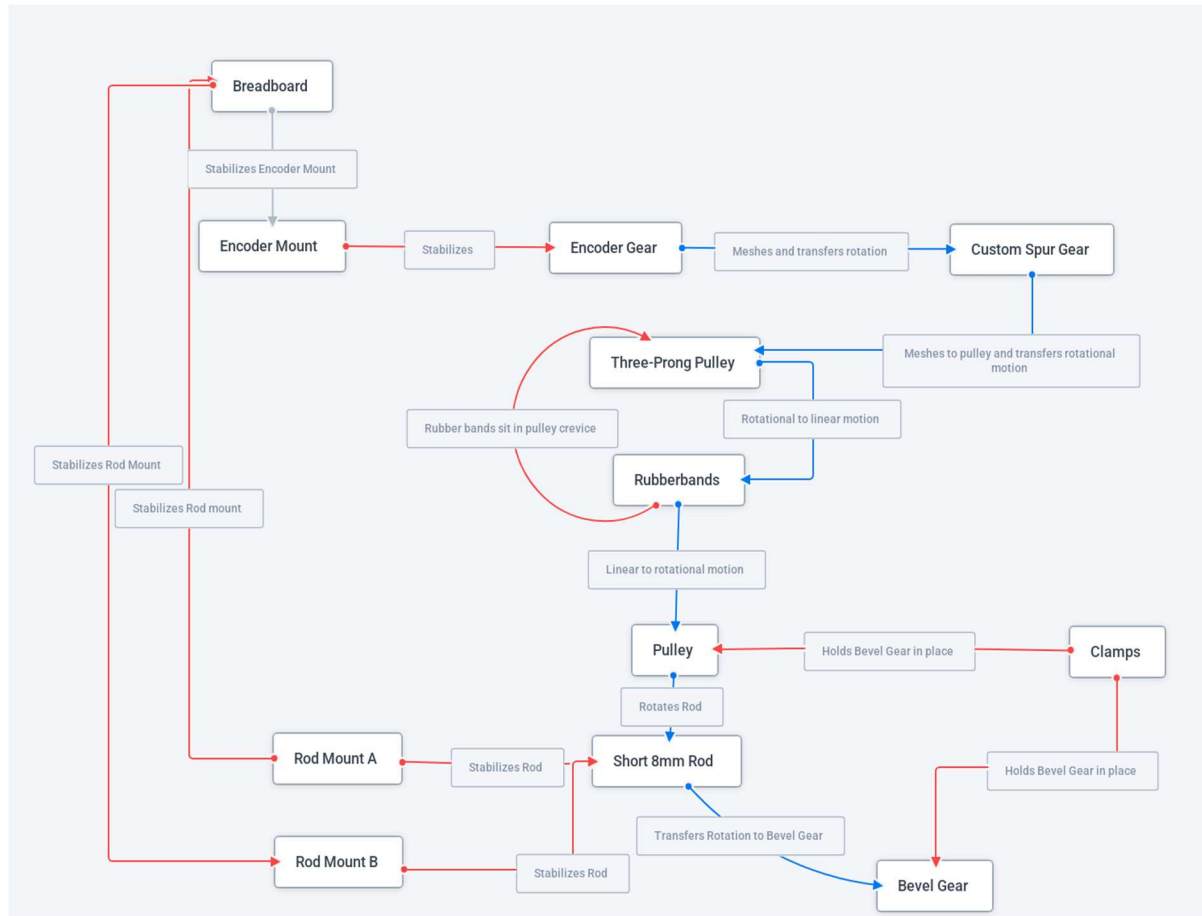


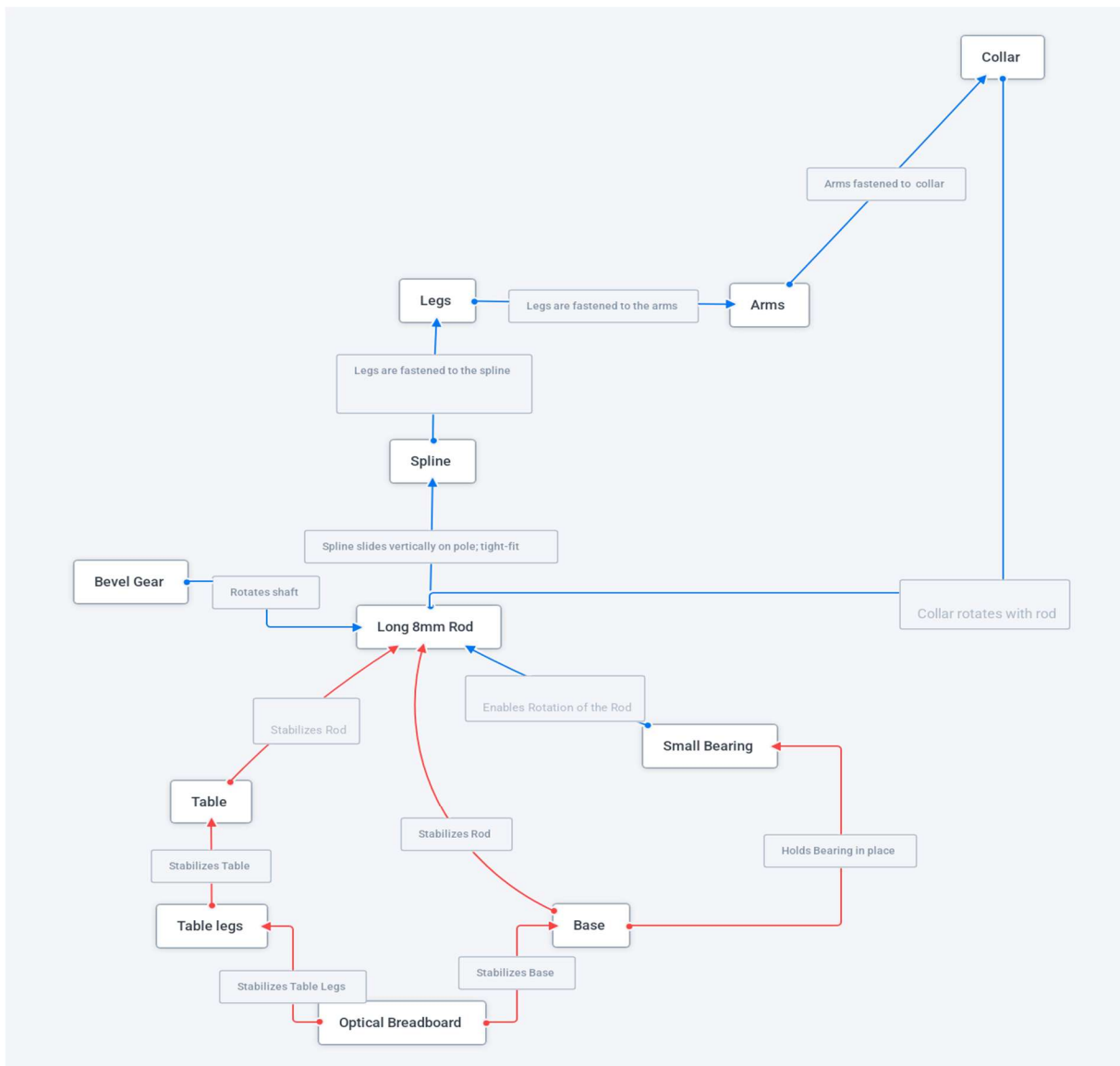
Analysis of the design:

Power Transfer:



To transfer the non-uniform motion to the system, the team decided on a pulley system. As the Encoder gear rotates, the three-pronged pulley rotates. This motion is transferred from the three-pronged pulley to a second pulley, which is attached to a Short Rod 8mm in diameter. Attached to the end of the rod is a bevel gear. As the rod rotates, the bevel gear rotates at the same rate, transferring motion by 90 degrees to the next subsystem.

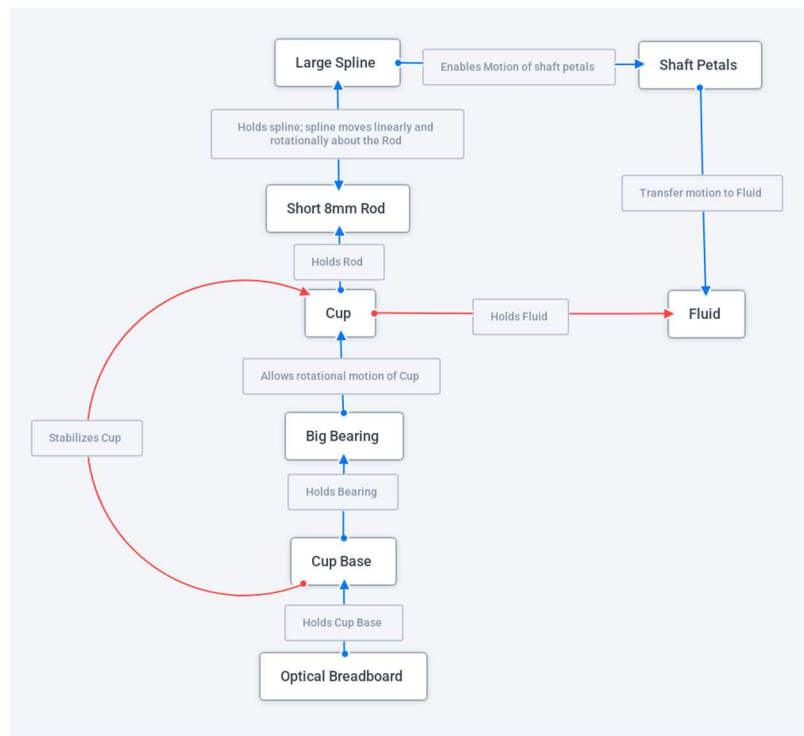
Centrifugal Governor:



The Centrifugal Governor arms hold weights and are attached to the Large 8mm Rod through the collar. The Large 8mm Rod rotates due to a bevel gear meshing to the previous subsystem. As the Large 8mm Rod rotates at a faster rate, the centrifugal force(force that causes something to spin in a circle) produced due to the mass of the weights, causes the arms to rise.

Since the motion from the encoder is non-uniform, the arms of the centrifugal governor effectively buff the change in rotational velocity. This vertical movement is captured through the legs and Small Spline of the centrifugal governor. Since the legs are connected to both the arms and the Small Spline, the vertical movement of the arms is transferred to the spline. As a result of this, the spline has linear and rotational motion along the Long 8mm Rod that it can transfer to the Fluid subsystem through rubber bands.

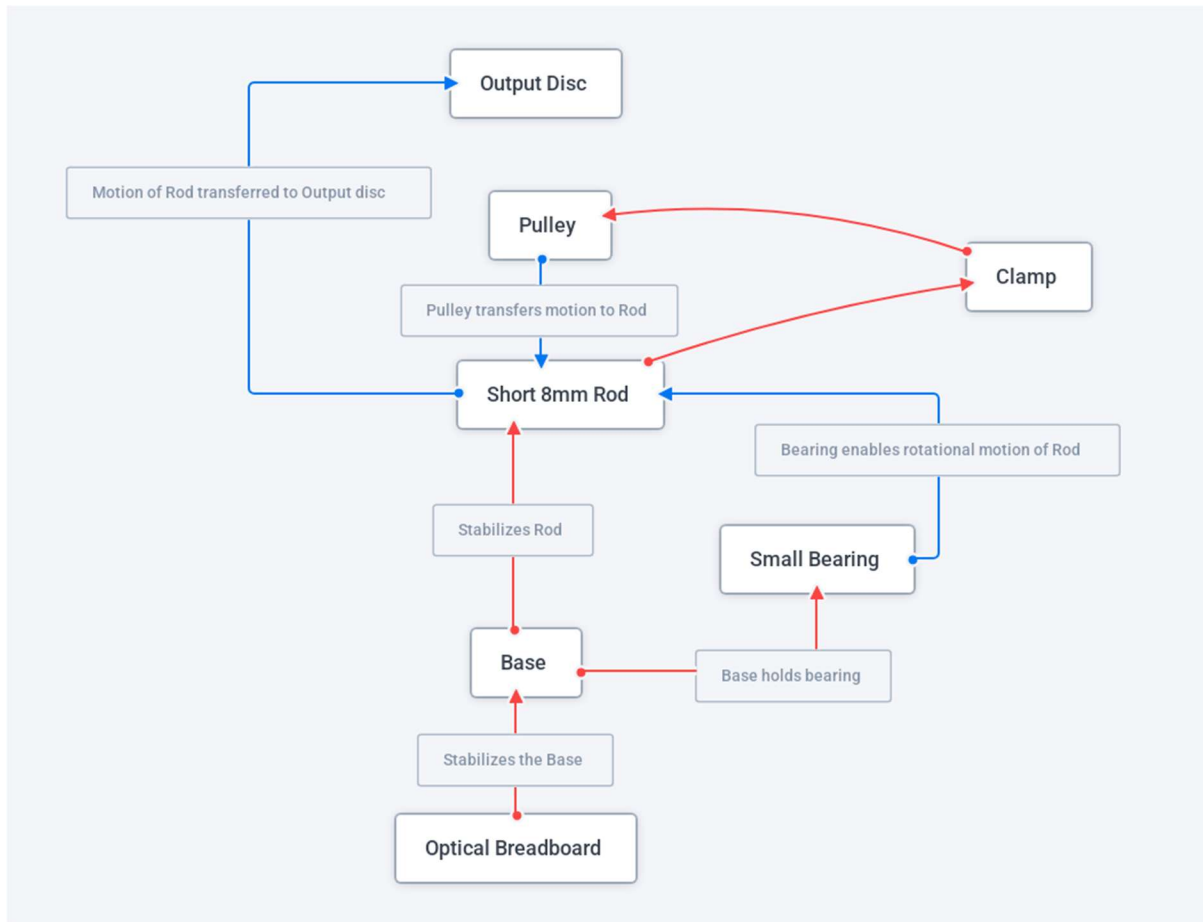
Fluid Subsystem:



Attached to the cup through a Short 8mm Rod is a Large Spline. Due to the spline in the previous subsystem, the Large Spline gains linear and rotational motion. Attached to the Large Spline are Petals which follow the same motion. The Petals convert rotational motion to the fluid. The momentum of the rotating fluid causes the Cup to rotate. Referring to previous subsystem

analysis, as the non-uniform motion changes, the number of petals in the fluid changes. Consequently, the rotational speed of the fluid and the cup change to account for the non-uniform motion. Connected to the Cup are rubber bands, which transfer the rotational motion of the Cup to the next subsystem.

Output:



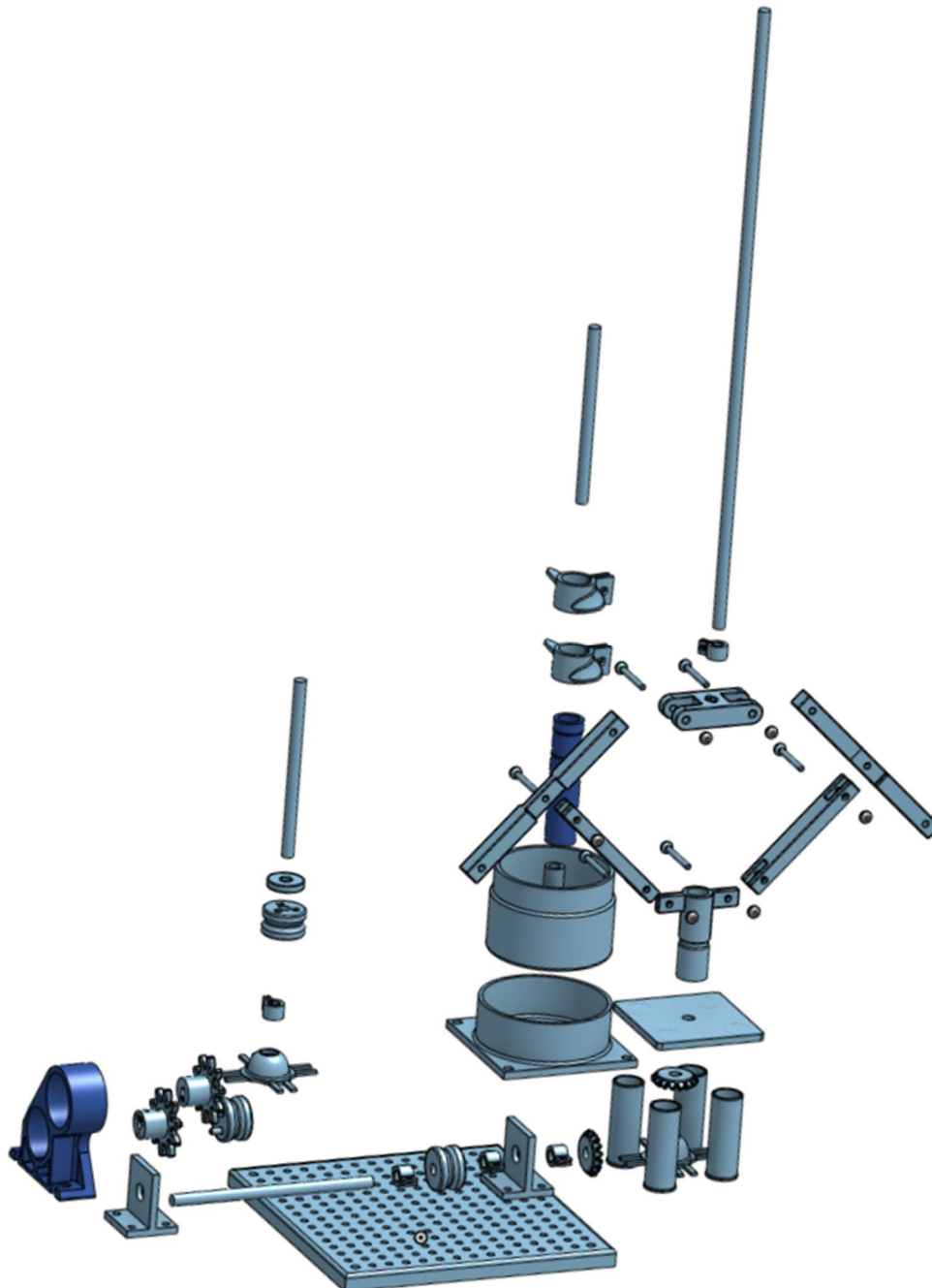
To transfer the rotational motion of the previous subsystem to the Output, rubberbands are placed on a pulley, connecting the Cup of the Fluid Subsystem to this subsystem. The pulley is

connected to a Short 8mm Rod to enable rotation. An output disc is attached to this rod to observe the final output of the Mechanical Rectifier.

Bill of Materials:

Component name	Quantity	Manufactured/Source	Cost	Brief Description
Arms	2	3D Print	N/A	Connect the centrifugal governor to the Small Spline
Base	2	3D Print	N/A	Stabilizes vertically placed rod components
Bearing big	3	Supplier(Cooper Union)	N/A	Enables rotation of Short 8mm Rod and Long 8mm Rod
Bearing small	1	Supplier(Cooper Union)	N/A	Enables rotation of Cup
Bevel Gear	2	3D Print	N/A	Transmits motion 90 degrees
Collar	1	3D Print	N/A	Attaches centrifugal governor to Long 8mm Rod
Cup	1	3D Print	N/A	Holds fluid and acts as a pulley
Cup Base	1	3D Print	N/A	Stabilizes the Cup
Custom Spur gear	1	3D Print	N/A	Meshes to the encoder gear
Fastener Pin	2	3D Print	N/A	Fastens weights onto Legs
Large Spline	1	3D Print	N/A	Enables rotational and linear motion of Shaft Petals
Legs	2	3D Print	N/A	Holds weights on ends; functions by centrifugal forces
Long 8mm Rod	1	Supplier(Cooper Union)	N/A	Enables motion
M3 Nuts	17	Supplier(Cooper Union)	N/A	Secures arms to legs, table legs to optical breadboard, arms to collar
M3 Screws	17	3D Print	N/A	Secures arms to legs, table legs to optical breadboard, arms to collar
M5 Screws	16	Supplier(Cooper Union)	N/A	Secures components to optical breadboard
Output disc	1	3D Print	N/A	Used to evaluate output motion
Pin Lock	2	3D Print	N/A	Secures Fastener Pins
Pulley	3	3D Print	N/A	Transfers rotational Power
Retaining Ring	5	3D Print	N/A	Secures bevel gear and pulley in power subsystem, pulley in output subsystem and
Rod Mount	2	3D Print	N/A	Secures Short 8mm Rod in Power Transfer subsystem
Shaft petals	2	3D Print	N/A	Cause fluid to rotate
Short 8mm Rod	3	Supplier(Cooper Union)	N/A	Enables motion of Power Transfer Subsystem and Output Subsystem
Small Spline	1	3D Print	N/A	Enables linear and rotational motion of the Fluid Subsystem
Table	1	3D Print	N/A	Stabilizes the Centrifugal Governor Subsystem
Table leg	4	3D Print	N/A	Stabilizes and attaches Table to Optical Breadboard
Three-Prong Pulley	1	3D Print	N/A	Meshes to the Custom Spur gear to transfer Encoder Gear motion
Washers	8	Supplier(Cooper Union)	N/A	Add weight to the Centrifugal Governor
RubberBand	9	Supplier(Cooper Union)	N/A	Connect pulleys of the Power Transfer Subsystem, Small Spline to Cup, Cup to Pulley in Output Subsystem
Optical Breadboard	1	Supplier(Cooper Union)	N/A	Acts as a base and stabilization for the whole system
Encoder Gear	1	Supplier(Cooper Union)	N/A	Provides non-uniform motion to the system

Exploded view of assembly:



Normal View of Assembly:

